

AI Engineering Master Study Schedule

A Custom, High-Yield Learning Blueprint for Split-Shift Professionals

WEEKLY WORK SCHEDULE CONSTRAINTS Weekdays: 08:00 - 11:00 & 16:26 - 23:00 Weekends: Shifts begin late at 16:18	TARGET STUDY ALLOCATION 11 Hours/Week (6 hrs Weekdays + 5 hrs Weekends) Uninterrupted Morning Devotions Honored
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The Master Weekly Time Architecture

This design anchors study sessions when mental clarity is highest, avoiding the fatigue of post-evening shift hours.

DAYS	TIME SLOT	DURATION	STRATEGIC FOCUS
Monday, Tuesday, Thursday, Friday	13:00 – 14:30	1.5 Hours / day	Mid-Day Focus: 30 mins reading theory/concepts, followed by 60 mins structured coding and execution.
Wednesday	—	0 Hours	Mid-Week Recovery: Absolute rest day. No study. Rest during the afternoon gap to reset cognitive load.
Saturday & Sunday	09:00 – 11:30	2.5 Hours / day	Deep Work Sprints: Morning exploration of heavy modules, deep architectural logic, and portfolio integration.

The 12-Month Curriculum Roadmap

A sequential progression moving steadily from programming syntax up to advanced machine learning systems design.

TIMELINE	CORE TEXT / RESOURCE	PRIMARY CONCEPTUAL FOCUS	PRACTICAL MILESTONE GOAL
Weeks 1–4			

TIMELINE	CORE TEXT / RESOURCE	PRIMARY CONCEPTUAL FOCUS	PRACTICAL MILESTONE GOAL
	PHASE 1 Python Crash Course	Syntax, OOP paradigms, file handling, testing.	Build a modular, clean command-line utility synced entirely to GitHub.
Weeks 5–8	PHASE 1 Grokking Algorithms	Arrays, lists, graphs, trees, search/sort efficiency (Big O).	Implement core search and sorting algorithms from memory in raw Python.
Weeks 9–14	PHASE 2 Mathematics for ML	Linear Algebra, Vector Calculus, Probability & Intro to Optimization.	Map out the geometric mechanics of Gradient Descent and Backpropagation.
Weeks 15–24	PHASE 2 Hands-On Machine Learning	Scikit-Learn pipeline creation, classical models, ensembles.	End-to-end Kaggle dataset regression/classification project with full EDA.
Weeks 25–30	PHASE 3 Deep Learning with PyTorch	Tensors, autograd, custom Datasets/ DataLoaders, CNNs.	Build, train, and validate a custom Convolutional Neural Network image classifier.
Weeks 31–36	PHASE 3 Build a Large Language Model (From Scratch)	Attention blocks, multi-head tokenization, transformers.	Code a small GPT-style transformer model and initialize pre-trained weights.
Weeks 37–42	PHASE 4 AI Engineering	Vector stores, Retrieval-Augmented Generation (RAG), orchestrators.	Construct a functional local document RAG assistant using FastAPI & Streamlit.
Weeks 43–48	PHASE 4 Designing Machine Learning Systems	Data pipelines, model monitoring, containerization, drift.	Wrap a trained model in Docker, serve via an API endpoint with basic health logging.

Detailed Week 1 Execution Plan

An exact day-by-day mapping for the commencement of your curriculum using *Python Crash Course*.

DAY	TIME BLOCK	DAILY TARGET TASKS
Monday	13:00 – 14:30	Environment Configuration: Download Python, configure VS Code. Initialize a structured repository on GitHub named <code>ai-engineering-journey</code> . Execute a simple <code>hello_world.py</code> .
Tuesday	13:00 – 14:30	Variables & Data Types: Read Chapter 2. Implement the "Try It Yourself" block exercises. Practice basic string operations (<code>.title()</code> , <code>.strip()</code>) and integers/floats. Push files to GitHub.
Wednesday	—	Mid-Week Rest Block: No study tasks. Intentionally disconnect from learning targets to secure sustainable long-term pacing.
Thursday	13:00 – 14:30	Intro to Lists: Read Chapter 3 (First Half). Code exercises managing elements within arrays. Experiment explicitly with <code>.append()</code> , <code>.insert()</code> , <code>.pop()</code> , and <code>del</code> commands.
Friday	13:00 – 14:30	Organizing Lists: Read Chapter 3 (Second Half). Work on sorting mechanisms. Contrast permanent array modifications using <code>.sort()</code> against temporary mutations using <code>sorted()</code> .
Saturday	09:00 – 11:30	Deep Work Block (Loops): Attack Chapter 4. Master the functional mechanics of <code>for</code> loops. Analyze indentation rules in Python and avoid scoping/syntax bugs.
Sunday	09:00 – 11:30	Synthesis & Portfolio Integration: Code a single robust script combining all Week 1 knowledge (create a dataset, loop through elements, mutate text, sort results). Add markdown comments, format clean code blocks, and push to GitHub.

Core Directives for Shift-Based Learning

- **Cloud Continuity:** Always maintain all scripts on GitHub or Google Colab. Transitioning seamlessly across hardware between shift intervals minimizes friction.

- **Cloud GPU Offloading:** During Phase 3 (PyTorch & LLMs), leverage free-tier Google Colab resources. Initialize extended training jobs right before leaving for your 16:26 evening shift.
- **Strict Evening Decompression:** Upon completing work at 23:00, avoid opening code editors or texts. Protect sleep quality to maintain optimal cognitive capacity during the next day's 13:00 learning window.